Week 8 reading guide and worked examples

Weekly overview

Required learning outcomes

By the end of this week, you should be able to:

- construct and interpret elementary confidence intervals for a population mean or proportion under stated normal, large-sample or Student's t conditions (S8); and
- formulate null and alternative hypotheses, calculate and interpret test statistics and p-values, distinguish one- and two-sided tests, and analyse Type I error, Type II error, power and test–confidence-interval links (S9).

Prerequisites and retrieval

Retrieve sampling distributions, standard errors, the law of large numbers and the central limit theorem (S6). Also retrieve the distinctions among a parameter, an estimand, an estimator and an observed estimate, together with bias, variance, mean squared error and consistency (S7).

The sheet integrates S7 retrieval in W08-Q01-PA. Its final exercise block then contains three further earlier-outcome retrieval parts: W08-Q10-PA (S6), W08-Q10-PB (S7) and W08-Q11-PA (M10, M11). These prepare you for the cumulative Statistics and Maths mock papers without adding a new Week 8 mathematics outcome.

For the newly taught outcomes, later sheet items retrieve methods after their initial teaching and practice: W08-Q06-PB and W08-Q09-PA revisit S8, while W08-Q09-PA revisits S9. The closed-note self-test then provides two further retrieval parts for each of S8 and S9.

Student materials

Before selecting or interpreting critical values, use the required distributions and critical values bridge note. It is point-of-need support within the existing reading and worked-example allocation, not additional workload.

1. Follow the exact written routes and skip instructions in the reading guide.
2. Study the original worked examples. Cover each solution and reproduce its main derivation from a blank page.
3. Attempt all questions in the exercise sheet without consulting full solutions.
4. Use the progressive hints only after a serious attempt.
5. Mark your work with the complete solutions. Record the first cause of each error by learning-outcome ID.
6. Complete the closed-note self-test.
7. Sit the separate 90-minute Maths and 90-minute Statistics mock papers under the conditions stated in the course syllabus.

No video, applet, code, calculator or specialist software is required. Every critical value, square-root approximation or normal probability needed for the intended hand calculation is supplied.

Suggested study sequence

1. Write the S6 and S7 prerequisite definitions from memory.
2. Read the confidence-interval selection first. Make a method card that distinguishes known-variance normal, unknown-variance normal, large-sample mean and large-sample proportion intervals.
3. Read the testing selection. Put the null distribution, tail convention and definition of a p-value beside every test.
4. Reproduce the institutional examples before attempting the sheet.
5. Complete Questions 1–4, then Questions 5–9, before the cumulative retrieval block.
6. Mark the sheet and self-test. Reattempt missed items from a blank page before sitting the mocks.
7. After marking both mocks, select no more than five outcome IDs for final targeted revision.

Workload and required outputs

The times below are provisional planning guidance, not a guarantee, and have not yet been verified by students. They are subject to first-cohort calibration. The Week 8 allocation is compressed because it also contains two timed mock papers.

Component	Target time	Required output
Required external reading, including its assigned source examples	2 h 40 min	A one-page method comparison for `S8` and `S9`
Institutional worked-example review and reproduction	50 min	All four examples studied; the central setup from at least one `S8` and one `S9` example reproduced
Week 8 problem sheet	4 h	All 17 scorable parts attempted and marked
Self-test, immediate marking and error-log update	30 min	Four self-test parts completed and diagnosed
Maths mock	1 h 30 min	Separate closed-note timed attempt
Statistics mock	1 h 30 min	Separate closed-note timed attempt
Core total	**11 h**	Completed Week 8 study and diagnostic cycle

The problem sheet contains 11 numbered questions, 17 scorable parts, 100 marks and a 240-minute target. Its difficulty balance is 30% foundation, 60% standard and 10% stretch. Parts whose primary demand is explanation, derivation or synthesis carry 58 marks.

The two timed mock sittings are included in the core total. Required post-attempt self-marking, diagnosis and spaced reattempt are separate provisional activities: no student-validated minute allowance is claimed, and they are excluded from the published core total pending first-cohort calibration.

End-of-course check

Before finishing, make sure that you can state why each interval or test is valid, not only substitute numbers into a formula. In particular, do not describe a p-value as the probability that the null hypothesis is true, and do not describe a realised confidence interval as assigning probability to a fixed parameter. Keep the error log for later targeted revision.

Required reading route

How to use this route

Complete the two required written selections in order. Open only the named sections and follow every skip instruction. The external selections remain on their authors' or institutions' sites; this guide does not reproduce their explanations, examples, questions or solutions.

Allow about 160 minutes for the required selections, including the named source examples. The separate institutional examples have their own 50-minute allowance for checking the course notation and distinctions. All times are approximate and student-unverified.

If formula rendering, PDF reading order or another source-format barrier prevents access to a required page, stop using that page. Use the original institutional worked examples, hints and complete solutions for the assessed method, and report the inaccessible locator so that an alternative can be provided. To report a broken, inaccessible

or unavailable required source, email cristiano.cantore@uniroma1.it with the week, item or source locator. You may request this confidentially without disclosing disability or other sensitive personal information. The programme Coordinator will provide an accessible replacement PDF or an approved alternative reading.

Required bridge note

Before selecting or interpreting critical values, read the distributions and critical values bridge note. It provides point-of-need distribution and critical-value support within the existing Week 8 reading and worked-example allocation, not additional workload.

1. Confidence intervals (`S8`)

Resource `STAT-S8-REQ-PISHRO`; source `PISHRO_NIK`. In Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*, read the following non-contiguous sections in order:

- §8.3.1, "The General Framework of Interval Estimation";
- §8.3.2, "Finding Interval Estimators";
- §8.3.3, "Confidence Intervals for Normal Samples"; and
- §8.3.4, "Solved Problems", restricted to the registered problems. Restrict the reading to confidence intervals for a population mean or proportion under normal, large-sample and Student's t conditions.

Within Section 8.3.4, use Solved Problems 2, 3, 5 and Problem 6(1) as written source examples. Re-derive the displayed interval setup before checking the source solution.

Skip: Section 8.3.4 Problem 1, Problem 4 and Problem 6(2), including confidence intervals for a variance. Do not use software or navigate into unassigned distribution or inference topics.

Target time: 70 minutes.

2. Hypothesis tests, errors, power and interval links (`S9`)

Resource `STAT-S9-REQ-MIT`; source `MIT_1805`. On Jeremy Orloff and Jonathan Bloom's MIT OpenCourseWare 18.05 classes and written-materials page, read:

- Class 17 Reading 17b, Sections 3–7.1;
- Class 18 Reading, Sections 3.1–5.4 except Section 5.3; and
- Class 23 Reading 23a, Sections 4 and 7.

Use only Problem Set 8 Problems 1(a–e), 2, 3(a–b(i–ii), c) and 4, and Problem Set 9 Problem 1(a–b), when following the registered written-practice route.

Skip: Class 18 Sections 5.3 and 5.5; Problem Set 8 Problems 1(f–g), 3(b)(iii–iv) and 5; and Problem Set 9 Problems 2–6. These skipped parts enter software, two-sample, Bayesian, chi-squared, F or ANOVA material outside this week's scope. Skip every linked video, applet and software activity.

Target time: 90 minutes.

Reading output

After each selection, close the source and write a brief original summary. Your one-page comparison should contain:

- the parameter or estimand;
- the estimator and its standard error;
- the assumptions or approximation conditions;
- the critical distribution and supplied critical value;
- the random interval or test statistic;
- the observed numerical conclusion; and
- one interpretation that is valid and one that is not.

Do not copy source prose, examples, exercise prompts or solutions into shared notes. The Week 8 institutional examples, questions and solutions are original course material.

Worked examples

These original institutional examples support `S8` and `S9`. Allow about 50 minutes to cover each **Solution**, follow the **Method**, and study all four examples. Reproduce the central setup from at least one `S8` and one `S9` example on a blank page. The Week 8 schedule is compressed by the two mock papers, so use these examples as concise notation and condition checks after the assigned reading.

Example 1 — a confidence interval for a mean (`S8`)

Suppose

$$X_1, \dots, X_{25} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 16),$$

where the population variance 16 is known. The observed sample mean is $\bar{x} = 42$. Construct a 95% confidence interval for the population mean μ. Use $z_{0.975} = 1.96$, where $\mathbb{P}(Z \leq z_{0.975}) = 0.975$ for $Z \sim \mathcal{N}(0, 1)$.

Method. Identify the fixed parameter, random estimator and observed estimate. Use the exact normal sampling distribution because the population is normal and its variance is known.

Solution. The parameter and estimand are both the population mean μ. The estimator is the random variable $\bar{X}$, and its observed estimate is $\bar{x} = 42$. The standard error is

$$\frac{\sigma}{\sqrt{n}} = \frac{4}{5} = 0.8.$$

The random 95% interval is

$$\left[\bar{X} - 1.96\frac{4}{5}, \bar{X} + 1.96\frac{4}{5}\right].$$

After observing the data, the realised interval is

$$[42 - 1.568,\ 42 + 1.568] = [40.432,\ 43.568].$$

Interpretation. In repeated independent samples, 95% of intervals constructed by this method cover the fixed parameter μ. After the sample is observed, this frequentist interval is fixed; it does not assign a 95% probability to the proposition that μ lies between these realised endpoints.

Example 2 — choosing Student's t (`S8`)

Suppose $X_1, \dots, X_9$ are independent observations from a normal population with unknown mean μ and unknown variance $\sigma^2 > 0$. The observed sample mean and sample standard deviation are $\bar{x} = 12$ and $s = 3$. Construct a 95% confidence interval for μ. Use $t_{0.975,8} = 2.306$.

Method. Because the population is normal, the variance is unknown and it is estimated by S^2, use the Student's t pivot with $n - 1 = 8$ degrees of freedom.

Solution. Under the stated assumptions,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_8.$$

The standard-error estimate is $s/\sqrt{n} = 3/3 = 1$. Therefore the realised 95% interval is

$$[12 - 2.306,\ 12 + 2.306] = [9.694,\ 14.306].$$

Interpretation. Replacing the unknown σ by S changes the reference distribution from standard normal to Student's t. A small sample does not by itself justify the t method; the normal-population condition is material here.

Example 3 — a one-sided p-value (S9)

Suppose

$$X_1, \ldots, X_{36} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 36),$$

with known variance. An observed sample has $\bar{x} = 102$. Test

$$H_0 : \mu = 100 \qquad \text{against} \qquad H_1 : \mu > 100$$

at significance level $\alpha = 0.05$. Use $\mathbb{P}(Z \geq 2) = 0.0228$ and $z_{0.95} = 1.645$.

Method. Standardise under the null value and use the upper tail selected by the alternative.

Solution. Under H_0,

$$Z = \frac{\bar{X} - 100}{6/\sqrt{36}} \sim \mathcal{N}(0, 1).$$

The observed statistic is

$$z_{\text{obs}} = \frac{102 - 100}{1} = 2.$$

The p-value is the null probability of a statistic at least as large as the observed value:

$$\mathbb{P}_{H_0}(Z \geq 2) = 0.0228.$$

Because $0.0228 < 0.05$, reject H_0. Equivalently, $2 > z_{0.95} = 1.645$.

Interpretation. The result is evidence against the null in the stated upper-tail direction. The p-value is not the probability that H_0 is true and does not measure the size or practical importance of the difference.

Example 4 — Type II error and power (S9)

Suppose a test of

$$H_0 : \mu = 10 \qquad \text{against} \qquad H_1 : \mu > 10$$

uses one observation $X \sim \mathcal{N}(\mu, 4)$ and rejects when $X > 13$. Find the Type I error probability. Then find the Type II error probability and power at the specific alternative $\mu = 14$. Use $\Phi(1.5) = 0.9332$ and $\Phi(-0.5) = 0.3085$, where $\Phi(z) = \mathbb{P}(Z \leq z)$.

Method. Evaluate the same fixed rejection rule under two different parameter values. Type I error is rejection under the null; Type II error is non-rejection at the specified alternative.

Solution. At the null value,

$$\alpha = \mathbb{P}_{\mu=10}(X > 13) = \mathbb{P}\left(Z > \frac{13-10}{2}\right) = 1 - \Phi(1.5) = 0.0668.$$

At $\mu = 14$, the Type II error probability is

$$\beta(14) = \mathbb{P}_{\mu=14}(X \leq 13) = \mathbb{P}\left(Z \leq \frac{13-14}{2}\right) = \Phi(-0.5) = 0.3085.$$

The power at $\mu = 14$ is

$$1 - \beta(14) = 0.6915.$$

Interpretation. The Type II error probability and power must name an alternative parameter value. They are not single properties of a composite alternative without such a value or a power function.

Reproduction check

Before starting the exercise sheet, reproduce:

1. one interval formula with its standard error and conditions; and
2. one test statistic with its null distribution, tail direction and decision rule.

If you cannot distinguish a parameter from an estimate, or a p-value from a posterior probability, return to the relevant example before continuing.